

ALBERT HALL AGCM, AB, Canada

WMO: 710050

Lat: 53.1134N

Lon: 111.1812W

Elev: 658

StdP: 93.67

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-32.8	-29.8	-36.3	0.1	-32.8	-33.0	0.2	-29.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	28.7	17.6	26.8	16.8	25.2	16.2	19.7	25.8	18.5	24.6	17.4	23.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	13.6	22.9	16.2	12.5	21.4	15.1	11.6	20.2	59.3	25.9	55.1	24.5	51.5	22.8	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-37.4	32.0	2.5	2.0	-39.2	33.4	-40.7	34.6	-42.0	35.8	-43.8	37.3
			-37.8	22.2	2.5	1.1	-39.6	23.0	-41.0	23.6	-42.4	24.2	-44.2	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-12.0	-13.0	-5.7	2.8	10.5	14.9	17.2	15.7	10.8	3.6	-5.3	-12.7
	DBStd	12.55	8.37	8.05	7.58	5.10	4.30	2.89	2.57	2.98	4.37	4.75	6.75	7.92
	HDD10.0	3489	682	644	486	220	47	3	0	1	41	204	460	703
	HDD18.3	5876	940	877	744	467	243	108	53	90	226	457	710	961
	CDD10.0	688	0	0	0	4	63	150	223	177	66	6	0	0
	CDD18.3	33	0	0	0	0	1	5	18	8	1	0	0	0
	CDH23.3	822	0	0	0	2	75	104	303	223	112	2	0	0
	CDH26.7	180	0	0	0	0	10	15	77	45	33	0	0	0
	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	414	19	12	16	30	44	70	76	58	34	18	19	17
	PrecMax	562	61	23	37	104	93	139	166	118	78	46	76	32
	PrecMin	209	4	3	5	3	5	14	25	12	1	2	2	4
	PrecStd	80	12	6	8	20	23	29	30	23	21	10	16	8
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	5.0	12.3	21.9	28.1	28.2	31.6	30.2	30.4	22.2	12.8	5.2
		MCWB	2.6	2.6	6.2	10.0	13.7	17.1	19.8	17.7	16.0	12.5	6.8	1.9
	2%	DB	4.1	2.8	8.6	18.3	25.6	26.1	28.7	27.6	26.6	17.1	7.7	2.8
		MCWB	1.6	0.7	4.2	8.4	12.7	15.7	19.5	17.6	14.6	9.6	3.8	0.3
	5%	DB	2.3	1.2	5.7	14.8	22.8	24.2	26.6	25.9	23.5	14.2	5.1	0.7
		MCWB	0.4	-0.2	2.7	6.8	11.6	15.2	18.5	17.5	13.6	7.8	2.3	-1.1
	10%	DB	0.1	-0.6	3.6	11.9	20.4	22.3	24.6	23.8	20.2	11.6	3.1	-1.6
		MCWB	-1.4	-1.8	1.3	5.4	10.5	14.2	17.5	16.4	12.4	6.6	1.1	-2.8
	0.4%	WB	2.9	2.7	7.2	10.7	14.7	18.9	22.0	20.8	17.1	12.8	7.2	2.2
		MCDB	5.2	4.7	11.8	21.7	25.2	25.0	28.5	26.3	25.9	20.8	11.6	4.8
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.9	1.1	4.5	8.7	13.4	17.3	20.3	19.2	15.5	10.5	4.3	0.6
		MCDB	4.0	2.3	8.2	17.0	22.6	23.5	26.6	25.2	25.0	15.2	7.0	2.6
	5%	WB	0.4	-0.1	2.8	7.2	12.4	16.2	19.2	17.9	14.1	8.8	2.7	-1.1
		MCDB	2.2	1.3	5.6	13.8	21.2	22.0	25.4	24.2	22.1	13.5	4.8	0.5
	10%	WB	-1.4	-1.9	1.4	5.8	11.3	15.1	18.0	16.8	12.9	7.1	1.2	-2.8
		MCDB	0.1	-0.6	3.4	11.3	19.3	20.6	23.5	22.4	19.7	11.4	2.9	-1.5
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.4	10.6	11.8	14.2	12.0	12.5	13.6	14.5	11.9	9.4	10.0
		MCDBR	11.6	11.6	12.6	18.0	18.7	15.7	16.7	18.2	21.4	17.1	11.9	12.0
		MCWBR	9.6	9.9	9.0	9.4	8.0	6.9	8.7	9.2	10.1	9.9	8.4	10.1
	5% WB	MCDBR	11.3	11.3	12.3	16.6	16.6	13.4	15.3	15.8	19.0	16.2	11.4	11.9
		MCWBR	9.5	9.8	8.9	9.0	7.5	6.7	8.3	8.6	9.5	9.6	8.3	10.2
(40) Clear-Sky Solar Irradiance	taub		0.251	0.292	0.294	0.333	0.353	0.372	0.365	0.379	0.315	0.279	0.252	0.226
	taud		2.317	2.224	2.371	2.355	2.348	2.328	2.371	2.312	2.464	2.541	2.448	2.362
	Ebn at Noon		753	809	901	900	896	878	877	841	867	841	759	731
	Edh at Noon		57	87	93	109	116	119	113	113	85	63	50	45
(44) All-Sky Solar Radiation	RadAvg		1.02	2.08	3.57	4.71	5.81	5.96	6.21	5.06	3.56	2.09	1.02	0.76
	RadStd		0.10	0.15	0.28	0.42	0.40	0.42	0.33	0.39	0.38	0.29	0.11	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page